

Containerization & DevOps Lab

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Experiment 11 — Orchestration using Docker Compose & Docker Swarm

Objective

To understand container orchestration and deploy a multi-container application using Docker Swarm. The experiment demonstrates scaling, load balancing, and self-healing capabilities compared with Docker Compose.

Theory

What is Container Orchestration?

Container orchestration is the **automatic management of containers**.

It handles tasks such as:

- scaling containers
- restarting failed containers
- distributing traffic across containers
- running services across multiple machines

An analogy is a **restaurant manager** who ensures enough staff are working, replaces unavailable staff, and distributes customers efficiently.

Progression of Container Management Tools

`docker run` → Docker Compose → Docker Swarm → Kubernetes

Tool	Purpose
docker run	run a single container
Docker Compose	run multiple containers together
Docker Swarm	orchestrate containers across nodes
Kubernetes	advanced orchestration platform

This lab focuses on **moving from Compose to Swarm orchestration**.

Prerequisites

- Docker installed
- Docker Swarm enabled
- WordPress + MySQL Compose file from Experiment 6

Docker Compose File

```
version: '3.9'

services:
  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
```

Task 1 — Stop Existing Containers

First ensure no containers are running.

```
docker compose down -v
docker ps
```

Expected result: no containers running.

Task 2 — Initialize Docker Swarm

Enable Swarm mode:

```
yuvraj_singh@LUCIF3R: exp6s docker swarm init --advertise-addr 172.22.87.136
Swarm initialized: current node (v6zzfufq8onqnlapf8k1oieez) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-1wnm13ttv0ooo7p2yy41blofcke95gv3ex0wh3eo55kagh9e9-6qsd51g19v34ta54kbinuu5p9 172.22.87.136:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

This command:

- enables swarm mode
- makes the current machine the **manager node**
- generates a join token for worker nodes

Verify swarm status:

```
yuvraj_chaula@LUCIF3R: exp6s docker node ls
ID                HOSTNAME        STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
v6zzfufq8onqnlapf8k1oieez * LUCIF3R        Ready     Active           Leader             29.4.0
```

Expected output shows the node status as **Ready / Leader**.

Task 3 — Deploy Application as a Stack

Deploy the compose file as a **stack**.

```
yuvraj_singh@LUCIF3R: lexp6s docker stack deploy ~C docker-compose.yml wpstack
Ignoring unsupported options: restart

Ignoring deprecated options:

container_name: Setting the container name is not supported.

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_db
Creating service wpstack_wordpress
```

Explanation:

Command	Meaning
docker stack deploy	deploy application in swarm
-c docker-compose.yml	specify compose file
wpstack	stack name

Behind the scenes Swarm:

1. reads the compose file
2. creates services
3. manages containers automatically.

Task 4 — Verify Deployment

List services:

```
yuvraj_chaula@LUCIF3R: exp6s docker service ls
ID                NAME                MODE                REPLICAS            IMAGE                PORTS
s0shrokwgbc1     wpstack_db          replicated          1/1                 mysql:5.7
t7fl7sa6r0t5     wpstack_wordpress   replicated          0/1                 wordpress:latest    *:8080->80/tcp
```

Example output:

```
ID NAME MODE REPLICAS IMAGE
xxxx wpstack_db replicated 1/1 mysql:5.7
xxxx wpstack_wordpress replicated 1/1 wordpress:latest
```

Inspect running tasks:

```
yuvraj_chamta@LUCIF3R: rexp6s docker service ps wpstack_worapress
ID                NAME                IMAGE                NODE                DESIRED STATE        CURRENT STATE        ERROR                PORTS
s88w74v0994n     wpstack_wordpress.1 wordpress:latest     LUCIF3R            Running              Running 3 seconds ago
```

Check containers:

```
yuvraj_singh@LUCIF3R:~/exp6s docker ps
CONTAINER ID        IMAGE                COMMAND              CREATED            STATUS              PORTS              NAMES
2faf08231f4a      wordpress:latest    "docker-entrypoint.s..." 24 seconds ago    Up 23 seconds      80/tcp             wpstack_wordpress.1.s8
8w74v0994ntwa4mitulwuvt
ccefbcf3aad3      mysql:5.7          "docker-entrypoint.s..." About a minute ago Up About a minute 3306/tcp, 33060/tcp wpstack_db.1.twogu8ihb
1gpe51wn65ivcw7r
```

Containers now have names like:

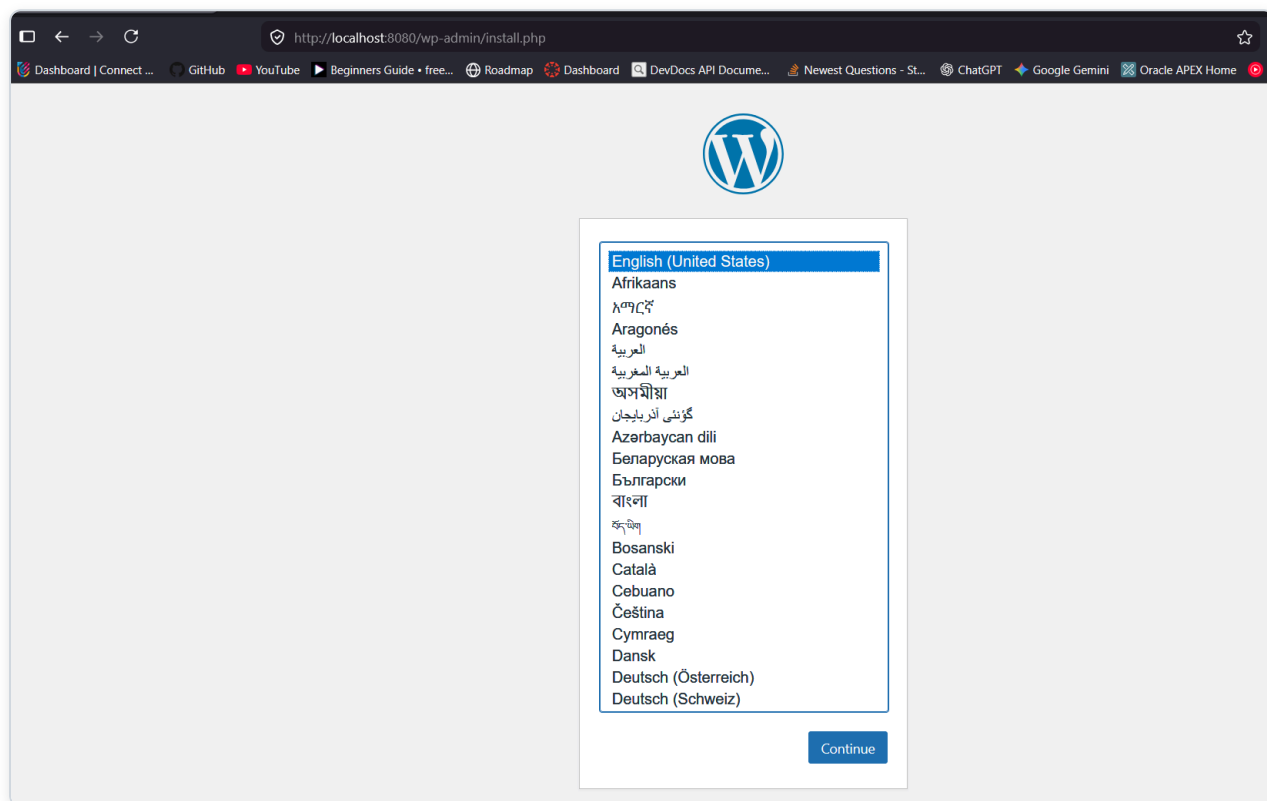
```
wpstack_wordpress.1.xxxx
wpstack_db.1.xxxx
```

Swarm now **manages containers automatically**.

Task 5 — Access the Application

Open browser:

```
http://localhost:8080
```



You should see the **WordPress setup page**.

Even though Swarm manages the containers, the application behaves the same.

Task 6 — Scale the Application

Scale WordPress service from **1 to 3 replicas**.

```
yuvraj_singh@LUCIF3R: 'exp6s docker service scale wpstack_wordpress-3
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running [=====>]
2/3: running [=====>]
3/3: running [=====>]
verify: Service wpstack_wordpress converged
```

Verify scaling:

```
yuvraj_chaula@LUCIF3R:~/exp6s$ docker service ls
```

ID	NAME	MODE	REPLICAS	IMAGE	PORTS
s0shrokwgbc1	wpstack_db	replicated	1/1	mysql:5.7	
t7f17sa6r0t5	wpstack_wordpress	replicated	3/3	wordpress:latest	*:8080->80/tcp

```
yuvraj_singh@LUCIF3R:~ /exp6s$ docker service ps wpstack_wordpress
```

ID	NAME	IMAGE	NODE	DESIRED STATE	CURRENT STATE	ERROR	PORTS
s88w74v0994n	wpstack_wordpress.1	wordpress:latest	LUCIF3R	Running	Running 7 minutes ago		
apw85cdg40or	wpstack_wordpress.2	wordpress:latest	LUCIF3R	Running	Running about a minute ago		
p2ct5h4opb1r	wpstack_wordpress.3	wordpress:latest	LUCIF3R	Running	Running about a minute ago		

```
yuvraj ff5ba9601355 chaula@LUCIF3R: wordpress exp6s Latest docker docker - ps --entrypoint grep wordpress 5= About minute ago Up About minute 80/tcp wpstack_wordpress.3.p2
```

CONTAINER ID	IMAGE	COMMAND	STATUS	STARTED AT	UP TIME	PORTS
ct0n4op01r0uaygnv0/n0zvt	wordpress:latest	"docker-entrypoint.s..."	Up	About a minute ago	About a minute	80/tcp
f8b6d250aad8	wordpress:latest	"docker-entrypoint.s..."	Up	7 minutes ago	7 minutes	80/tcp
w85cdg40or1b0b5lrcwepz1	wordpress:latest	"docker-entrypoint.s..."	Up	7 minutes ago	7 minutes	80/tcp
2faf08231f4a	wordpress:latest	"docker-entrypoint.s..."	Up	7 minutes ago	7 minutes	80/tcp
8w74v0994ntwa4m1tulwuvt	wordpress:latest	"docker-entrypoint.s..."	Up	7 minutes ago	7 minutes	80/tcp

Now **three WordPress containers run simultaneously.**

Load Balancing

Even with multiple containers, users still access:

```
http://localhost:8080
```

Swarm creates an **internal load balancer** that distributes traffic across all containers automatically.

Task 7 — Test Self-Healing

Swarm automatically replaces failed containers.

Find a WordPress container:

```
docker ps | grep wordpress
```

Kill it:

```
docker kill <container-id>
```

Check service tasks:

```
yuvraj_singh@LUCIF3R:~/exp6s$ docker service ps wpstack_worapress
ID                NAME                IMAGE                NODE                DESIRED STATE       CURRENT STATE          ERROR                PORTS
s88w74v0994n     wpstack_wordpress.1  wordpress:latest    LUCIF3R             Running             Running 11 minutes ago
apw85cdg40or     wpstack_wordpress.2  wordpress:latest    LUCIF3R             Running             Running 5 minutes ago
p2ct5h4opb1r     wpstack_wordpress.3  wordpress:latest    LUCIF3R             Running             Running 5 minutes ago
```

Swarm detects the failure and **creates a replacement container automatically**.

Task 8 — Remove the Stack

Clean up all services:

```
yuvraj_singh@LUCIF3R: exp6s$ docker stack rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
```

Verify removal:

```
yuvraj_chamta@LUCIF3R: exp6s$ docker service ls
ID                NAME                MODE                REPLICAS            IMAGE                PORTS
```

Containers should be removed.

Compose vs Swarm

Feature	Docker Compose	Docker Swarm
Scope	single host	multi-node cluster
Scaling	basic	built-in service scaling
Load balancing	no	yes
Self-healing	no	yes
Rolling updates	no	yes
Service discovery	container names	DNS + VIP

Compose is best for **development**, while Swarm supports **production orchestration**.

Observations

1. The same Compose YAML file can be reused in Swarm.
 2. In Swarm, you manage **services instead of containers**.
 3. Swarm resolves port conflicts using an internal load balancer.
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Learning Outcome Questions

1. Why is Docker Compose not sufficient for production environments?
 2. What does `docker stack deploy` do differently from `docker compose up` ?
 3. How does Swarm provide self-healing?
 4. What happens if a container managed by Swarm is killed?
 5. Can the same Compose file be used for development and production?
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Conclusion

This experiment demonstrated container orchestration using Docker Swarm.

A multi-container WordPress application was deployed using a Compose file, scaled to multiple replicas, and tested for self-healing capabilities. Swarm provides built-in load balancing, automatic recovery, and cluster management, making it more suitable for production environment

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